Rishika Patel

+91 7985925729 | rishika.patel2419@gmail.com | linkedin.com/in/rishika-patel | github.com/Rishikaaz

SUMMARY

A motivated Computer Science student specializing in Cybersecurity and Ethical Hacking with hands-on experience in Vulnerability Assessment, Penetration Testing, and Malware Analysis. Proven technical lead skilled in deploying SIEM solutions using the ELK Stack and engineering machine learning models to solve security challenges.

TECHNICAL SKILLS

Programming & Scripting: C++, Python, Bash Scripting, C, SQL, Linux Shell Scripting
Security & Ethical Hacking: Computer Networking, Vulnerability Assessment & Penetration Testing (VAPT), Reverse Engineering, Malware Analysis, Threat Detection, Web Security, OWASP Top 10, Network Reconnaissance
Security Tools & Frameworks: BurpSuite, Wireshark, Nmap, Metasploit, Nessus, Splunk, Ghidra, Scapy, Kali Linux, Hashcat
Cloud, DevOps & Infrastructure: AWS (EC2, S3, IAM, VPC), Docker Compose, Linux Administration, ELK Stack
Networking & Protocols: TCP/IP, OSI Model, Socket Programming, Network Traffic Analysis, Packet Capture

PROJECTS

Containerized HIDS & SIEM Pipeline | *Python, Docker, ELK stack, Scapy, Nmap | Cybersecurity Project* ***May 2026***

- Spearheaded a containerized SIEM architecture using Docker Compose, optimizing host CPU/RAM overhead by reducing resource consumption by 40% through strategic VM migration.
- Developed a Python HIDS sensor via Scapy to capture live network traffic, stream JSON alerts over TCP, and validate behavioral rules using simulated Nmap reconnaissance attacks.

Vulnerability Management Platform | *Python, SQLite, Network Security, Socket Programming | Cybersecurity Project* ***Jul 2026***

- Engineered a modular vulnerability management platform in Python automating network reconnaissance, asynchronous port scanning, CVE signature matching, and HTTP header auditing.
- Integrated SQLite database schemas to track, correlate, and index system vulnerabilities over time, cutting manual reporting effort by 50%.
- Enhanced an automated security reporting engine that aggregates structured database findings into executive-level vulnerability assessment reports.

AI-Powered Behavioral Anomaly Detection |*Python, Scikit-learn, Pandas, Scapy, Networking | ML & Cybersecurity Project* ***Jul 2026***

- Designed an end-to-end threat detection pipeline that captures raw network packets using Scapy, converts them into behavioral features, and passes them to unsupervised models (Isolation Forest, Autoencoders).
- Trained the system on baseline network traffic to identify subtle zero-day anomalies and multi-stage attack patterns that traditional signature-based rules miss.
- Streamed real-time JSON alert logs directly into SIEM workflows, cutting down false positives and enabling immediate incident response by 60%.

EXPERIENCE

Cyber Security and Ethical Hacking Intern | Tamizhan Skills **Jun – Jul 2025** ***Remote***

- Programmed 6+ proof-of-concept Python security tools, including a custom multi-port socket scanner, file-encryption utilities, and AES-encrypted client-server communication channels.

Green Skills with AI Technologies Intern | Edunet Foundation **Jun – Jul 2025** ***Remote***

- Engineered data preprocessing and feature selection pipelines in Python for environmental datasets, reducing model training variance and improving report generation efficiency by 25%.

EDUCATION

Vellore Institute of Technology | *B.Tech in Computer Science and Engineering (CGPA: 8.84/10)* **Bhopal, India** ***2023 – 2027***

St. Xavier's Convent School | *Senior Secondary (XII), ICSE* **Lucknow, India** ***2022***

St. Xavier's Senior Secondary School | *High School (X), CBSE (91.4%)* **Gonda, India** ***2020***

CERTIFICATIONS

- Certified Cybersecurity Educator Professional (CCEP)
- Blockchain and its Applications (NPTEL, IIT Kharagpur)
- Networking Basics & Cyber Threat Management (CISCO)
- The Bits and Bytes of Computer Networking (Coursera)
- AWS Cloud Practitioner & AWS Skill Builder
- Cybersecurity Job Simulation (Mastercard & Deloitte)

ACHIEVEMENTS

- Solved 300+ LeetCode problems and actively competed in Codeforces contests with a max of 902 contest rating.
- Selected for the DSCI Cyber For Her Hackathon, qualified for Round 2 of Scripted By{Her} 2.0, Meesho Hackathon as well as in Honeywell and the APCSIP-2026 for a Cyber Cell internship.
- Ranked in the top 2% globally on TryHackMe with a 177-day learning streak in penetration testing and digital forensics.
- Placed 355th out of 5,947 global competitors in the TryHackMe Industrial-Intrusion CTF.
- Secured 7th rank nationally in the Shell n'Zen CTF, solving complex web security and reverse engineering challenges.